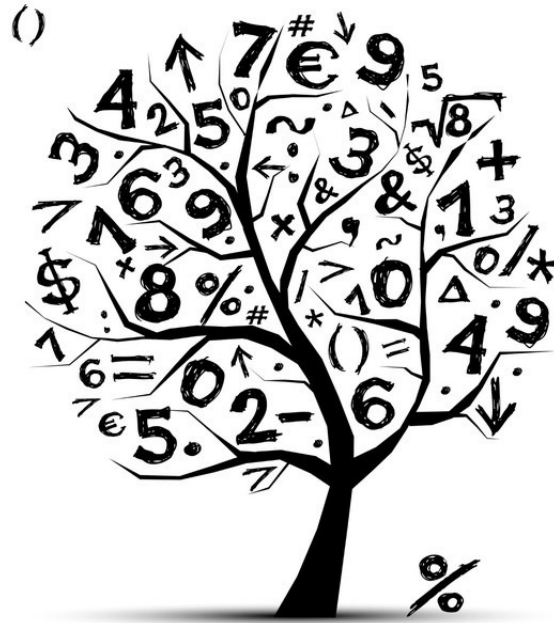


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Exercises: products



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Exercise 1

Compute:

a) $5!$, b) $\frac{7!}{5!}$, c) $\frac{10!}{7!}$, d) $\frac{10!}{7!3!}$

a) $5! = \prod_{j=1}^5 j = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$.

b) Using the formula $\frac{n!}{(n-k)!} = \prod_{j=1}^k (n-j+1)$ with $n = 7$ and $k = 2$: $\frac{7!}{5!} = 7 \cdot 6 = 42$.

c) With $n = 10$ and $k = 3$: $\frac{10!}{7!} = 10 \cdot 9 \cdot 8 = 720$.

d) From c): $\frac{10!}{7!3!} = \frac{720}{3!} = \frac{720}{6} = 120$.

Exercise 2

Prove that for every natural number $n \geq 1$ the product of the first n positive even numbers is:

$$\prod_{j=1}^n 2j = 2^n n!$$

and compute its value for $n = 5$.

Using the product by a constant (with $c = 2$ and $a_j = j$) we have:

$$\prod_{j=1}^n 2j = 2^n \prod_{j=1}^n j = 2^n n!$$

For $n = 5$: $\prod_{j=1}^5 2j = 2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 = 2^5 \cdot 5! = 32 \cdot 120 = 3840$.

Exercise 3

Prove that for every natural number $n \geq 1$ the product of the first n odd numbers is:

$$\prod_{j=1}^n (2j-1) = \frac{(2n)!}{2^n n!}$$

and compute its value for $n = 4$.

The factors of $(2n)! = \prod_{j=1}^{2n} j$ are the n even numbers $2, 4, \dots, 2n$ and the n odd numbers $1, 3, \dots, 2n-1$. Rearranging the factors and using Exercise 2, we have:

$$(2n)! = \prod_{j=1}^n 2j \prod_{j=1}^n (2j-1) = 2^n n! \prod_{j=1}^n (2j-1)$$

and then

$$\prod_{j=1}^n (2j-1) = \frac{(2n)!}{2^n n!}$$

For $n = 4$: $1 \cdot 3 \cdot 5 \cdot 7 = 105$ and indeed $\frac{8!}{2^4 \cdot 4!} = \frac{40320}{16 \cdot 24} = \frac{40320}{384} = 105$.

Exercise 4

- a) Prove that for every sequence of non-zero real numbers $a_1, a_2, \dots, a_{n+1}$ we have (**telescoping product**):

$$\prod_{j=1}^n \frac{a_{j+1}}{a_j} = \frac{a_{n+1}}{a_1}$$

- b) Using a), compute $\prod_{j=1}^n \frac{j+1}{j}$.

- a) Using the product of products, the index translation and the decomposition, we have:

$$\prod_{j=1}^n \frac{a_{j+1}}{a_j} = \frac{\prod_{j=1}^n a_{j+1}}{\prod_{j=1}^n a_j} = \frac{\prod_{j=2}^{n+1} a_j}{\prod_{j=1}^n a_j} = \frac{\left(\prod_{j=2}^n a_j \right) a_{n+1}}{a_1 \prod_{j=2}^n a_j} = \frac{a_{n+1}}{a_1}$$

- b) Applying a) with $a_j = j$:

$$\prod_{j=1}^n \frac{j+1}{j} = \frac{n+1}{1} = n+1$$

Exercise 5

Prove that for every natural number $n \geq 2$ we have:

$$\prod_{j=2}^n \left(1 - \frac{1}{j^2} \right) = \frac{n+1}{2n}$$

and compute its value for $n = 10$.

Since

$$1 - \frac{1}{j^2} = \frac{j^2 - 1}{j^2} = \frac{(j-1)(j+1)}{j^2} = \frac{j-1}{j} \frac{j+1}{j}$$

from the product of products we have:

$$\prod_{j=2}^n \left(1 - \frac{1}{j^2}\right) = \prod_{j=2}^n \frac{j-1}{j} \prod_{j=2}^n \frac{j+1}{j}$$

Both products are telescoping (Exercise 4, with the index starting from 2): with $a_j = \frac{1}{j}$ we have $\frac{a_j}{a_{j-1}} = \frac{j-1}{j}$, and with $a_j = j$ we have $\frac{a_{j+1}}{a_j} = \frac{j+1}{j}$. Then:

$$\prod_{j=2}^n \frac{j-1}{j} = \frac{1}{2} \cdot \frac{2}{3} \cdots \frac{n-1}{n} = \frac{1}{n}, \quad \prod_{j=2}^n \frac{j+1}{j} = \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = \frac{n+1}{2}$$

and therefore

$$\prod_{j=2}^n \left(1 - \frac{1}{j^2}\right) = \frac{1}{n} \frac{n+1}{2} = \frac{n+1}{2n}$$

For $n = 10$ we get $\frac{11}{20}$.

Exercise 6

Using the index translation and the index reflection, prove that:

$$\text{a) } \prod_{j=3}^n (j-2) = (n-2)! \quad (n \geq 3), \quad \text{b) } \prod_{j=0}^{n-1} (n-j) = n! \quad (n \geq 1)$$

and compute the product in a) for $n = 8$.

- a) With the index translation by $m = 2$ (setting $i = j - 2$, when j goes from 3 to n , i goes from 1 to $n - 2$):

$$\prod_{j=3}^n (j-2) = \prod_{i=1}^{n-2} i = (n-2)!$$

For $n = 8$: $\prod_{j=3}^8 (j-2) = 6! = 720$.

- b) With the index reflection (with $a_j = j$, so that $a_{n-j} = n - j$):

$$\prod_{j=0}^{n-1} (n-j) = \prod_{j=0}^{n-1} a_{n-j} = \prod_{j=1}^n a_j = \prod_{j=1}^n j = n!$$

Exercise 7

Prove that for every natural number $n \geq 1$ we have:

$$\prod_{j=1}^n 2^j = 2^{\frac{n^2+n}{2}}$$

and compute its value for $n = 4$.

Since the product of powers with the same base is the power with the sum of the exponents ($2^a 2^b = 2^{a+b}$), we have:

$$\prod_{j=1}^n 2^j = 2^1 2^2 \dots 2^n = 2^{1+2+\dots+n} = 2^{\sum_{j=1}^n j} = 2^{\frac{n^2+n}{2}}$$

For $n = 4$: $2^1 \cdot 2^2 \cdot 2^3 \cdot 2^4 = 2^{10} = 1024$.

Exercise 8

Prove that for every natural number $n \geq 1$ we have:

$$2^{n-1} \leq n! \leq n^n$$

We use the fact that if $0 \leq a_j \leq b_j$ for every $j \in \{1, \dots, n\}$, then $\prod_{j=1}^n a_j \leq \prod_{j=1}^n b_j$ (the inequalities between non-negative numbers can be multiplied member to member).

- Upper bound: since $j \leq n$ for every $j \in \{1, \dots, n\}$, using the product with constant term we have:

$$n! = \prod_{j=1}^n j \leq \prod_{j=1}^n n = n^n$$

- Lower bound: for $n = 1$ we have $2^0 = 1 = 1!$. For $n \geq 2$, using the decomposition and since $j \geq 2$ for every $j \in \{2, \dots, n\}$, we have:

$$n! = 1 \cdot \prod_{j=2}^n j \geq \prod_{j=2}^n 2 = 2^{n-1}$$

Exercise 9

Prove that $(n+1)! = (n+1) n!$ for every natural number $n \geq 0$, and use it to simplify, for $n \geq 1$, the expression

$$\frac{(n+1)! - n!}{(n-1)!}$$

Then compute its value for $n = 6$.

For $n = 0$ we have $1! = 1 = 1 \cdot 0!$. For $n \geq 1$, from the decomposition:

$$(n+1)! = \prod_{j=1}^{n+1} j = \left(\prod_{j=1}^n j \right) (n+1) = (n+1) n!$$

Then, using it twice ($n! = n (n-1)!$ holds for $n \geq 1$):

$$\frac{(n+1)! - n!}{(n-1)!} = \frac{(n+1) n! - n!}{(n-1)!} = \frac{n n!}{(n-1)!} = \frac{n n (n-1)!}{(n-1)!} = n^2$$

For $n = 6$: $\frac{7! - 6!}{5!} = \frac{5040 - 720}{120} = \frac{4320}{120} = 36 = 6^2$.